1 Introduction

How do the opinions of a group evolve when its members listen to each other, and what changes when some of them resist? Models of opinion dynamics describe a social network as a weighted digraph, whose nodes are the agents and whose edges carry the influence of one agent on another, and update the opinions by repeated averaging. This report follows the tutorial by Proskurnikov and Tempo [5], the course slides [2, 3, 1] and, as a support, the book by Bullo [4].

The historical account of [5] explains why the report compares two models. The first model of this kind was proposed by French in 1956: every agent replaces its opinion with the mean of the opinions displayed to it, and French used it to measure the *social power* of each agent, i.e. its ability to influence the common opinion of the group; his conditions for consensus were proved rigorously by Harary in 1959 and 1965. In 1974 DeGroot extended the model to an arbitrary stochastic matrix, as a procedure to reach a consensus of subjective probabilities: this is the *French–Harary–DeGroot* (FDG) model. In 1964 Abelson had proposed its continuous-time counterpart and observed that consensus, the outcome of a very broad class of such models, is too strong to be realistic: what must be assumed to explain the persistent disagreement observed in real groups? This question, the *community cleavage problem*, points to agents that resist the others. In 1968 Taylor extended Abelson's model with constant external opinions, the *prejudices*, to which the agents remain attached; its discrete-time counterpart came only in the 1990s, with the *Friedkin–Johnsen* (FJ) model, in which every agent mixes the average of the opinions it listens to with a fixed prejudice, and which has been validated experimentally on small and medium-size groups.

The thread of the report is stubbornness: an agent that never changes its opinion, and the graded mechanisms of which it is the extreme case, a self-weight in the FDG model and a prejudice in the FJ model. We ask how these mechanisms change three things: whether the opinions converge, what they converge to, and how much each agent weighs in the result, measured by the social power in the FDG model and by the influence centrality in the FJ model. The two models are compared on the same small networks ~~, with~~ and with the same resisting agents, agent 1 ~~as the agent that resists~~in most examples, so that every difference comes from the model and not from the data~~.~~; an auxiliary network, used only for the FJ model, shows what happens when a prejudice does not reach every agent. The continuous-time models of Abelson and Taylor are mentioned only as an overview and are not studied in this report.

Section 1.1 recalls the notions on graphs and non-negative matrices used throughout. Section 2 introduces the FDG model and studies stubbornness and social power in it; Section 3 does the same for the FJ model, from its convergence and the augmented graph to the influence centrality; Section 4 compares the two models.

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides [2, 3, 1].

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \dots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides [1]; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction [5].

Definition 1.1 (Weighted adjacency matrix [2])**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n the (weighted) adjacency matrix of G is the $n \times n$ matrix A in which:

(i) for each edge $(i, j) \in E$ the (i, j) entry of A is equal to the edge weight a_{ij},

(ii) all other entries of A are equal to zero.

Definition 1.2 (Connectivity of digraphs [2])**.** A directed graph $G = (V, E)$ is

(i) *strongly connected* if there is a directed walk from any node to any other node;

Social power. The entry w_i is called the *social power* of agent i: it is the weight of the initial opinion $x_i(0)$ in the final opinion of the group, i.e. in the consensus value [5]

$$\lim_{k\to\infty} x_i(k) = w^\top x(0) = \sum_{j=1}^{n} w_j\, x_j(0) \qquad \text{for every agent } i. \tag{2}$$

This definition requires a consensus: by (A2), all the rows of $\lim_{k\to\infty} A^k = \mathbf{1}_n w^\top$ are equal to $w^\top$, so a single vector describes the whole group. Without a consensus, as in Theorem 2.3, the rows of the limit matrix differ, each agent ends at its own weighted average of the initial opinions, and no single vector plays the role of w.

Point (i) has a direct social interpretation: the consensus value is the same for every agent, so the initial opinion of agent i can enter it only if it reaches *all* the agents, i.e. if i is globally reachable. Hence the consensus value depends only on the initial opinions of the globally reachable agents.

Among the globally reachable agents, the social power depends on their position in the network, and it can be regarded as a centrality measure [5]. For instance, in the model of French with $n = 3$ agents and

$$A = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/3 & 1/3 & 1/3 \\ 0 & 1/2 & 1/2 \end{bmatrix},$$

all agents are globally reachable, but agent 2, the only one listened to by all the others, has the largest social power, $w = (2/7,\, 3/7,\, 2/7)^\top$ [5].

2.1.3 Disagreement

Theorem 2.1 requires a single globally reachable group, i.e. a unique sink in the condensation digraph $C(G)$. We now consider the case in which $C(G)$ has $n_s \geq 2$ sinks. Each sink is a strongly connected component, so its periodicity is well defined (Definition 1.3).

Theorem 2.3 (Disagreement in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. Assume that the condensation digraph $C(G)$ has $n_s \geq 2$ sinks. The following statements are equivalent:*

(A1) the eigenvalue 1 is semi-simple with multiplicity n_s and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent;

(A3) each sink of $C(G)$ is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then the solution of the DT averaging system satisfies

$$\lim_{k\to\infty} x(k) = v \neq \alpha \mathbf{1}_n, \quad \alpha \in \mathbb{R}.$$

The final opinions can be described group by group. The agents of a sink are not influenced by anyone outside it, so each sink evolves as an FDG model on its own and, being aperiodic, reaches an internal consensus that depends only on the initial opinions of its members; every agent outside the sinks converges to a convex combination of the consensus values of the sinks [1]. In social terms, a sink is a closed group that listens only to itself, and the agents in between settle on an intermediate opinion: the disagreement is bounded by the smallest and the largest consensus value of the sinks.

2.1.4 Rosetta stone

Table 1, the so-called *Rosetta stone* [1], summarizes all the possible behaviours of the FDG model: it translates each property of the row-stochastic matrix A into a property of the digraph G and into the asymptotic outcome, with an example for each case; the translation between A and G comes from Theorems 1.12 and 1.13 in cases (i)–(ii), and from Theorems 2.1 and 2.3 in cases (iii)–(iv). In case (i) both agents are globally reachable and A is doubly-stochastic, so by Corollary 2.2 the group agrees on the average of the initial opinions. In case (iii) agent 1 ~~is stubborn (Definition 2.4~~listens only

- $t = 1$: agent 1 is stubborn and is the only globally reachable node, so $w = e_1$: the stubborn agent takes all the social power, and the initial opinions of the other agents have no effect on the final opinion of the group.

Between these two extremes, w is the left eigenvector of $A(t)$ associated with the eigenvalue 1, normalized to sum 1 (condition (A2) of Theorem 2.1)~~; Figure 3 reports it for $t \in (0, 1]$. Numerically, the social power of agent 1 grows with t towards~~. On both networks it is

$$w(t) = \frac{1}{1 + 5(1-t)} \big(1,\ 1-t,\ 1-t,\ 1-t,\ 1-t,\ 1-t\big)^\top,$$

as one checks by substituting $w = (w_1, b, \ldots, b)$ in $w^\top A(t) = w^\top$: in both networks agents $2, \ldots, 6$ receive the same total weight from the others, and the equations reduce to $b = (1-t)\, w_1$. Hence $w_1 = 1/(1 + 5(1-t))$ grows from $1/6$ to 1, ~~while~~ the other agents share the rest equally (~~$w_1 \approx 0.286$ and $w_i \approx 0.143$~~ $w_1 = 2/7$ and $w_i = 1/7$ for $i \neq 1$ when $t = 1/2$), ~~with the same values on the two networks. The bottom panel shows where the group converges,~~ and the consensus value ~~$w^\top x(0)$~~ (point (ii) of Theorem 2.1) ~~: it~~ is

$$w(t)^\top x(0) = \frac{3(1-t)}{6 - 5t},$$

which moves from the average 0.5 of the initial opinions, for ~~small t~~ $t \to 0$, to $x_1(0) = 0$ for $t = 1$, and ~~it is ≈ 0.429~~ is $3/7 \approx 0.4286$ for $t = 1/2$, the value reached in Figure 2~~; also this curve is~~. Figure 3 plots the two curves, which are the same on the two networks. In other words, on these two networks the self-weight of agent 1 changes how the opinions converge, which depends on the network, but not where they converge.

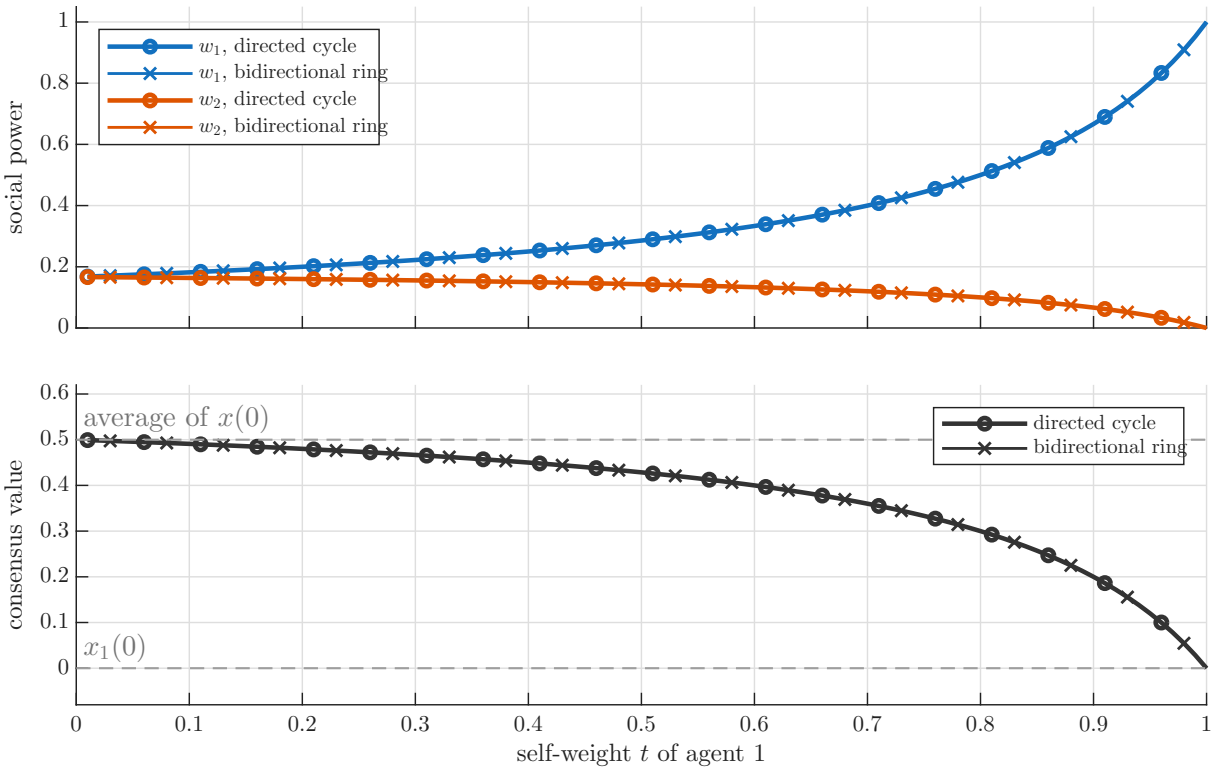


Figure 3: Top: social power of agent 1 (w_1) and of agent 2 (w_2) as a function of the self-weight $t \in (0, 1]$ of agent 1. Bottom: consensus value $w^\top x(0)$ for $x(0) = (0,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 1)^\top$; the dashed lines mark the average of $x(0)$ and $x_1(0)$. Both panels are computed numerically in the directed cycle (dots) and in the bidirectional ring (crosses); the two networks give the same values.

With stubborn agents, the FDG model converges under the following condition [5].

Corollary 2.5 ([5]). *Let the group have $s \geq 1$ stubborn agents, influencing all other individuals (i.e. every other node of G is connected by a walk to the set of stubborn nodes). Then the FDG model (1) is convergent.*

Remark 2.6 (More than one stubborn agent). If two or more agents are stubborn, each of them is a sink of $C(G)$ made of a single, aperiodic node. If every other agent is influenced, directly or through intermediaries, by at least one stubborn agent, these are the only sinks~~, and by Theorem 2.3~~: the group converges ~~, but~~ (Corollary 2.5), but by Theorem 2.3 to a disagreement: every stubborn agent keeps its own initial opinion, and every other agent converges to a convex combination of the opinions of the stubborn agents it is influenced by (see also [5]). Since there is no single consensus value, the social power is no longer defined. This situation, in which several agents are anchored to different fixed opinions, is the starting point of the Friedkin–Johnsen model of Section 3.

3 The Friedkin–Johnsen model

3.1 Introduction to the FJ model

3.1.1 The FJ model

In the FDG model every agent keeps averaging the opinions it listens to, and a persistent disagreement can only come from the structure of the network, i.e. from several closed groups (Theorem 2.3). The Friedkin–Johnsen (FJ) model adds a second ingredient: each agent can also be tied to a fixed opinion of its own, called its *prejudice*. Following [5], the model reads

$$x(k+1) = \Lambda W x(k) + (I - \Lambda)\, u, \tag{4}$$

where

- $W \in \mathbb{R}^{n\times n}$ is a row-stochastic matrix, the *matrix of social influences*, i.e. the weighted adjacency matrix of the influence digraph G (Definition 1.1), which plays the role of A in the FDG model;
- $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$ the *susceptibility* of agent i to social influence;
- $u \in \mathbb{R}^n$ is a constant vector collecting the agents' *prejudices*.

Componentwise,

$$x_i(k+1) = \lambda_i \sum_{j=1}^{n} w_{ij}\, x_j(k) + (1 - \lambda_i)\, u_i. \tag{5}$$

At every step agent i averages the opinions it listens to, exactly as in the FDG model, and mixes this average with its prejudice: the weight λ_i goes to the others and the complement $1-\lambda_i$ goes to u_i. The two quantities play different roles: the prejudice u_i is an opinion and says *where* the agent is anchored, while $1-\lambda_i$ is a weight and says *how strongly*; we call $1-\lambda_i$ the *attachment* of agent i to its prejudice.

The special cases of (4) are the following [5].

- If $\lambda_i = 1$, agent i follows the FDG rule. If $\Lambda = I$, the FJ model reduces to the FDG model with $A = W$; for this reason we assume $\Lambda \neq I$ from now on.
- If $\lambda_i < 1$, agent i is called *prejudiced*: it is anchored to its prejudice u_i.
- If $\lambda_i = 0$, then $x_i(k) = u_i$ for all $k \geq 1$: the agent ignores the others and jumps to its prejudice after one step. Its opinion never changes, i.e. the agent is stubborn in the sense of Definition 2.4, only if in addition $u_i = x_i(0)$. A stubborn agent of the FJ model is therefore an agent with $\lambda_i = 0$ and $u_i = x_i(0)$.

In social terms, the self-weight of the FDG model and the prejudice of the FJ model are two different forms of resistance to influence. A self-weight ties an agent to its *current* opinion, which still moves towards the others (Section 2.2.1); a prejudice ties it to a *fixed* opinion, which never moves.

In the original formulation of Friedkin and Johnsen the prejudices coincide with the initial opinions, $u = x(0)$, and the susceptibilities are coupled to the self-weights through $1 - \lambda_i = w_{ii}$ [5]. As in [5], we do not impose these conditions: u and $x(0)$ are independent, and so are Λ and W. The choice $u = x(0)$ will be needed in Section 3.1.4 to define the influence centrality.

and $\rho(\Lambda^{11}W^{11}) < 1$ by Theorem 3.2. Hence the eigenvalues on the unit circle, which decide convergence and asymptotic stability (Theorem 1.9), can only come from W^{22}, i.e. from the FDG model of the P-independent agents: the model is asymptotically stable when there are no P-independent agents ($r = n$), and it converges when the eigenvalue 1 of W^{22} is semi-simple and its other eigenvalues lie inside the unit disk, i.e. when W^{22} is regular.

3.1.3 How the opinions converge: the matrix V and the augmented graph

When all agents are P-dependent, the final opinions are given by (9), and the convex combination of Theorem 3.2 involves only the prejudices:

$$x_i(\infty) = \sum_{j=1}^{n} v_{ij}\, u_j.$$

Three facts follow.

- The initial opinions are forgotten: $x(0)$ does not appear in $x(\infty)$.
- Only the prejudices of prejudiced agents count: if $\lambda_j = 1$, the j-th column of $I - \Lambda$ is zero, hence so is the j-th column of V, and u_j has no weight in any final opinion.
- Every final opinion lies between the smallest and the largest prejudice of the prejudiced agents.

The comparison with the FDG model is instructive. There, the consensus value $w^\top x(0)$ is a weighted average of the initial opinions, with the same weights w for every agent (Theorem 2.1); here, the final opinions are weighted averages of the prejudices, and each agent has its own weights, given by its row of V.

In the FDG model, consensus is a property of the model: the model reaches a consensus if the final opinions coincide for any initial condition $x(0)$ [5]. In the FJ model with all agents P-dependent, the limit $x(\infty) = Vu$ does not depend on $x(0)$ (equation (9)): whether the final opinions coincide is decided by the prejudices u, which are an external input, through the matrix V, and no longer by the initial opinions. Accordingly, [5] studies the FJ model in terms of convergence and of the form of the limit, and not in terms of consensus.

For a given prejudice vector u, we still say that the FJ model reaches a *consensus* if the final opinions coincide, $x(\infty) = c\,\mathbf{1}_n$, and a *disagreement* otherwise. When all agents are P-dependent, a consensus is reached if and only if all the prejudiced agents have the same prejudice. Indeed, if $x(\infty) = c\,\mathbf{1}_n$, the equation (5) of a prejudiced agent i at the equilibrium reads $c = \lambda_i c + (1-\lambda_i)u_i$, i.e. $(1-\lambda_i)(c-u_i) = 0$, hence $u_i = c$. Conversely, if $u_i = c$ for every prejudiced agent, $c\,\mathbf{1}_n$ is an equilibrium, since the rows of W sum to 1, and by asymptotic stability it is the only one. The prejudices of the agents with $\lambda_i = 1$ play no role, since their columns of V are zero. Section 3.2.2 shows both cases on the same networks: a single prejudice, on which the group agrees, and different prejudices, which keep it divided.

The notion of consensus of the FDG model applies again only to the P-independent agents ($0 < r < n$), which evolve as an FDG model with matrix W^{22} and whose limit depends on their initial opinions $x^2(0)$ (Section 3.1.2). The P-dependent agents converge to convex combinations of the prejudices and of the final opinions of the P-independent agents (Theorem 3.2), so they do not necessarily end up at the consensus value of the P-independent agents, even when there is one.

The limit (9) has a direct interpretation in terms of the FDG model. Following [5], the FJ model (4) may be considered as an augmented FDG model with n virtual stubborn agents, anchored at the prejudices: the virtual agent i' associated with agent i keeps the opinion $x_{i'}(k) = u_i$ for all k. Explicitly, stacking the opinions x of the real agents and x' of the virtual agents,

$$\begin{bmatrix} x(k+1) \\ x'(k+1) \end{bmatrix} = \begin{bmatrix} \Lambda W & I-\Lambda \\ 0 & I \end{bmatrix} \begin{bmatrix} x(k) \\ x'(k) \end{bmatrix}, \qquad x'(0) = u. \tag{11}$$

Row i of the matrix in (11) sums to $\lambda_i + (1-\lambda_i) = 1$, and row i' sums to 1; hence the matrix is row-stochastic and (11) is an FDG model on $2n$ agents. In its digraph, which we call the *augmented graph*:

Two descriptions of the same dynamics. When all the agents are P-dependent, the FJ model is asymptotically stable (Corollary 3.3), while the augmented model (11), whose matrix is row-stochastic and has the eigenvalue 1 (Theorem 1.7), is never asymptotically stable: it converges to the disagreement of case (iv), which depends on the initial opinions $x'(0) = u$ of the virtual agents. The two statements describe the same trajectories and do not contradict each other. The matrix $\bar{A}$ of (11) is block triangular, so

$$\operatorname{spec}(\bar{A}) = \operatorname{spec}(\Lambda W) \cup \{\underbrace{1, \dots, 1}_{n}\}.$$

The eigenvalues of ΛW lie inside the unit disk, and the FJ model, which has only these, is asymptotically stable. The augmented model adds the eigenvalue 1, once for each virtual agent and semi-simple by Theorem 2.3 applied to the n aperiodic sinks of the augmented graph: these are the prejudices, which in the augmented model are states, and they place it in case (iv) of Table 1. The difference lies in what is a state. In the FJ model u is an input: asymptotic stability concerns the perturbations of $x(0)$ with u fixed, which die out as in (10), while changing u is not a perturbation but a different input, with a different equilibrium Vu. In the augmented model u is part of the state: changing it is a perturbation of the initial condition along the directions of the eigenvalue 1, and it is never forgotten; perturbing only the real agents, with $x'(0)$ unchanged, the augmented model forgets the perturbation exactly as the FJ model. Section 3.2.1 reads the perturbation experiment of Figure 7 in these terms.

3.1.4 Influence centrality

In the FDG model the social power w is defined only when the group reaches a consensus (Section 2.1.2), which is not the relevant notion for the FJ model (Section 3.1.3). Nor does the augmented FDG model (11) help: its graph has at least n sinks, so for $n \geq 2$ it never reaches a consensus and its social power is not defined (Section 3.2.3 discusses the special case of a single prejudice). A different measure is needed, and [5] discusses an extension of the social power to the FJ model, introduced by Friedkin.

Influence centrality. As in [5], we confine ourselves to the asymptotically stable FJ model with $u = x(0)$, the traditional assumption of the Friedkin–Johnsen theory (Section 3.1.1); then $x(\infty) = Vx(0)$ by (9). The assumption $u = x(0)$ is what gives the initial opinions a role: in general $x(\infty) = Vu$ does not depend on $x(0)$, so the initial opinions would have no weight in the final opinions. The social power of agent i is generalized as the mean weight of its initial opinion in determining the group members' final opinions [5]. These mean weights are the elements of the non-negative vector c such that

$$c = n^{-1} V^\top \mathbf{1}_n, \qquad \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i(\infty) = \frac{1}{n} \mathbf{1}_n^\top V x(0) = c^\top x(0), \tag{12}$$

and c_i is called the *influence centrality* of agent i [5]. In terms of V, c_i is the average of the i-th column of V, whose entries are the weights of $x_i(0)$ in the final opinions of the single agents. Since V is stochastic, c is non-negative and $c^\top \mathbf{1}_n = 1$, like w (Theorem 2.1): both distribute a total weight 1 among the agents. Similar to French's social power, the influence centrality is generated by an opinion formation mechanism [5].

The relation with the social power is made precise in [5]. The convergence and the consensus of the FDG model are equivalent, respectively, to the regularity and the full regularity of the stochastic matrix A [5] (Definition 1.10). For a fixed W, let $\Lambda = \alpha I_n$ with $\alpha \in (0, 1)$, i.e. the same susceptibility for all agents, and consider the corresponding matrix V_α and vector c_α. For $\alpha = 0$, i.e. $\Lambda = 0$, every agent keeps its initial opinion, $V = I$ and $c_0 = n^{-1}\mathbf{1}_n$: the social power is uniformly distributed between all agents [5].

Lemma 3.5 ([5]). *Let W be a fully regular matrix and w stand for the vector of French's social power, corresponding to the FDG model* (1) *with $A = W$. Then, $w = \lim_{\alpha \to 1-0} c_\alpha$.*

The limit cannot be replaced by $\alpha = 1$: for $\Lambda = I$ the FJ model is the FDG model, which is not asymptotically stable, and $I - W$ is singular since $1 \in \operatorname{spec}(W)$ (Theorem 1.7), so V is not defined. In social terms, the influence centrality moves from the uniform distribution, when every agent is anchored to its own initial opinion, to French's social power, when the weight of the prejudices vanishes.

t of Section 2.2: this choice makes the two parameters directly comparable, both equal to 1 in the stubborn case.

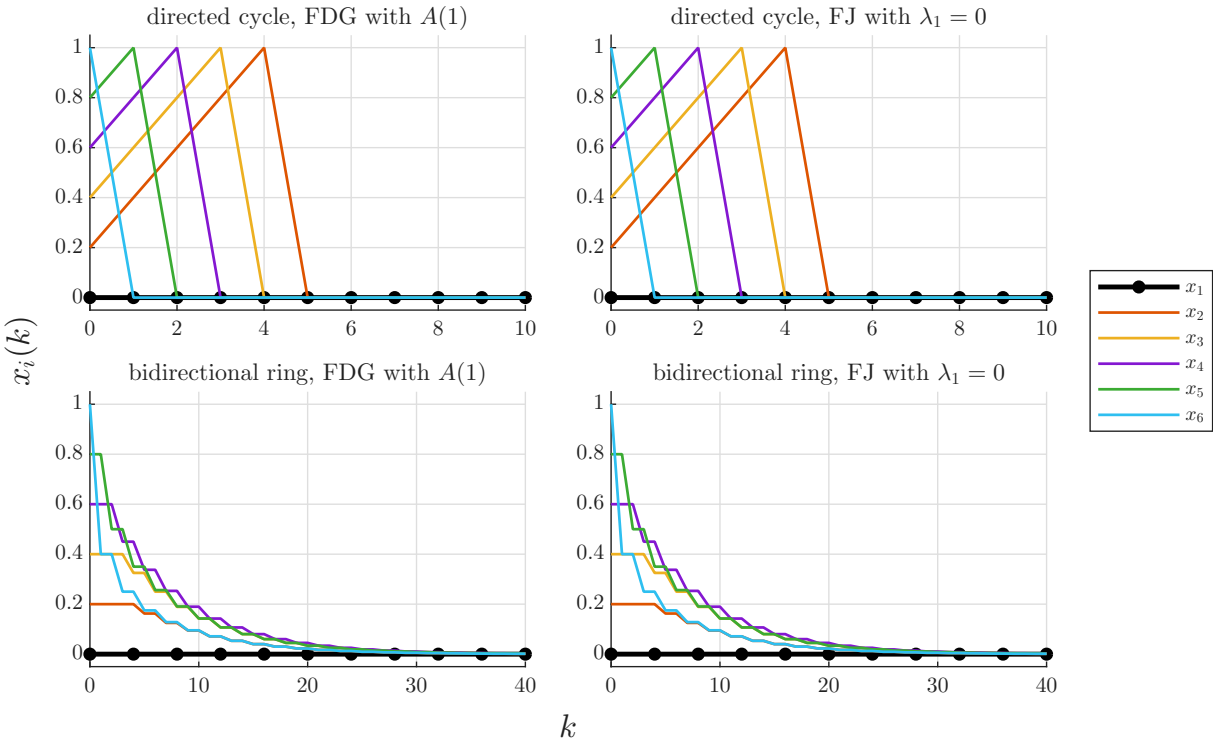


Figure 6: Opinions $x_i(k)$ in the FDG model with $A(1)$ (left) and in the FJ model with $\lambda_1 = 0$, $\lambda_i = 1$ for $i \neq 1$ and $u = x(0)$ (right), on the directed cycle ($K = 10$, top) and on the bidirectional ring ($K = 40$, bottom), with $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$. Agent 1 is black and marked with dots.

State or input. The two models of Figure 6 have the same trajectories, but not the same stability. ~~The FDG model with stubborn agents converges under the following condition.~~

~~Let the group have $s \geq 1$ stubborn agents, influencing all other individuals (i.e. every other node of G is connected by a walk to the set of stubborn nodes). Then the FDG model (1) is convergent.~~

Under the assumption of Corollary 2.5, every agent of the FJ model obtained by the rewriting is prejudiced or influenced by a prejudiced agent, so the FJ model is asymptotically stable (Corollary 3.3), whereas the FDG model is convergent but not asymptotically stable, since a row-stochastic matrix always has the eigenvalue 1 (Theorem 1.7, see also [5]). The explanation given in [5] is that changing the initial condition $x(0)$ of the FJ model leaves the prejudice vector u unchanged, while in the FDG model the prejudice is part of the state vector.

Figure 7 starts the two models again from $x_1(0) = 0.5$, leaving u unchanged: now $u_1 = 0 \neq x_1(0)$, and agent 1 of the FJ model is prejudiced but no longer stubborn.

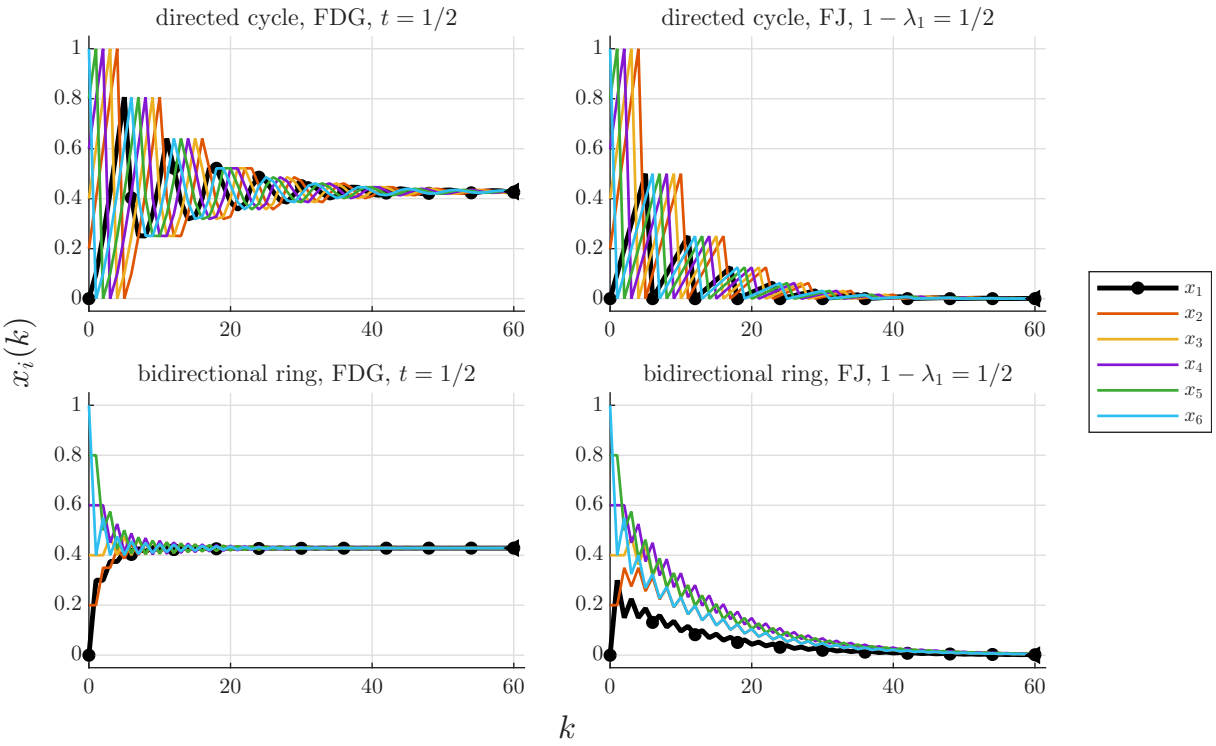


Figure 9: Opinions $x_i(k)$ with the same weight $1/2$ on agent 1, with $x(0) = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$ and $K = 60$. Left: FDG model with self-weight $t = 1/2$ (matrix $A(1/2)$, current opinion); right: FJ model with $\lambda_1 = 1/2$, $\lambda_i = 1$ for $i \neq 1$ and $u = x(0)$ (attachment $1 - \lambda_1 = 1/2$, initial opinion). Top: directed cycle; bottom: bidirectional ring. The triangles mark the limits: the consensus value ≈ 0.4286 of the FDG model and $u_1 = 0$ for the FJ model.

Both networks are strongly connected, so every agent is connected by a walk to agent 1: all the agents are P-dependent and, by Corollary 3.3 (case $\Lambda \neq I_n$, W irreducible), the model is asymptotically stable, although the same networks oscillate forever in the FDG model without self-loops. Since column j of V is zero when $\lambda_j = 1$ (Section 3.1.3) and V is stochastic (Theorem 3.2), $V = \mathbf{1}_6 e_1^\top$ and $x(\infty) = u_1 \mathbf{1}_6 = 0$ for every attachment $1 - \lambda_1 > 0$: the group reaches a consensus on the prejudice of agent 1, and crosses the FDG compromise 0.4286 on its way. ~~The equilibrium shows why: agents $2, \ldots, 6$ copy, or average, the agents they listen to, so at an equilibrium they all take the value of agent 1, whose equation becomes $x_1^* = \lambda_1 x_1^* + (1 - \lambda_1) u_1$, i.e. $(1 - \lambda_1)(x_1^* - u_1) = 0$. The~~ This is the condition of Section 3.1.3 with a single prejudiced agent: the fixed value u_1 makes the equilibrium unique, and the attachment only changes how fast it is reached; in the FDG model, instead, every consensus is an equilibrium, and the initial opinions decide which one is reached. In the augmented graph (Section 3.1.3) the only virtual agent that the real agents reach, directly or through intermediaries, is $1'$, which plays the role of the stubborn agent of Section 2.2.1: the whole group converges to its value u_1. In social terms, a self-weight is inertia, which slows an agent down but lets it follow the group, while a prejudice is an anchor: a single anchored agent, however loosely, decides where the whole group ends.

Every agent prejudiced. In Figure 10 every agent resists the influence of the others with the same weight 0.2, in both models: in the FDG model (left) as a self-weight on its current opinion, $A = 0.2\, I + 0.8\, W$; in the FJ model (right) as an attachment to its initial opinion, $\Lambda = 0.8\, I$ and $u = x(0)$. The triangles mark the limits.

On the directed cycle,

$$A = 0.2\,I + 0.8\,W = \begin{bmatrix} 0.2 & 0.8 & 0 & 0 & 0 & 0 \\ 0 & 0.2 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.8 & 0 \\ 0 & 0 & 0 & 0 & 0.2 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0 & 0.2 \end{bmatrix}, \qquad \Lambda W = 0.8\,W = \begin{bmatrix} 0 & 0.8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}:$$

A is row- and column-stochastic, with a positive diagonal, while every row of ΛW sums to 0.8.

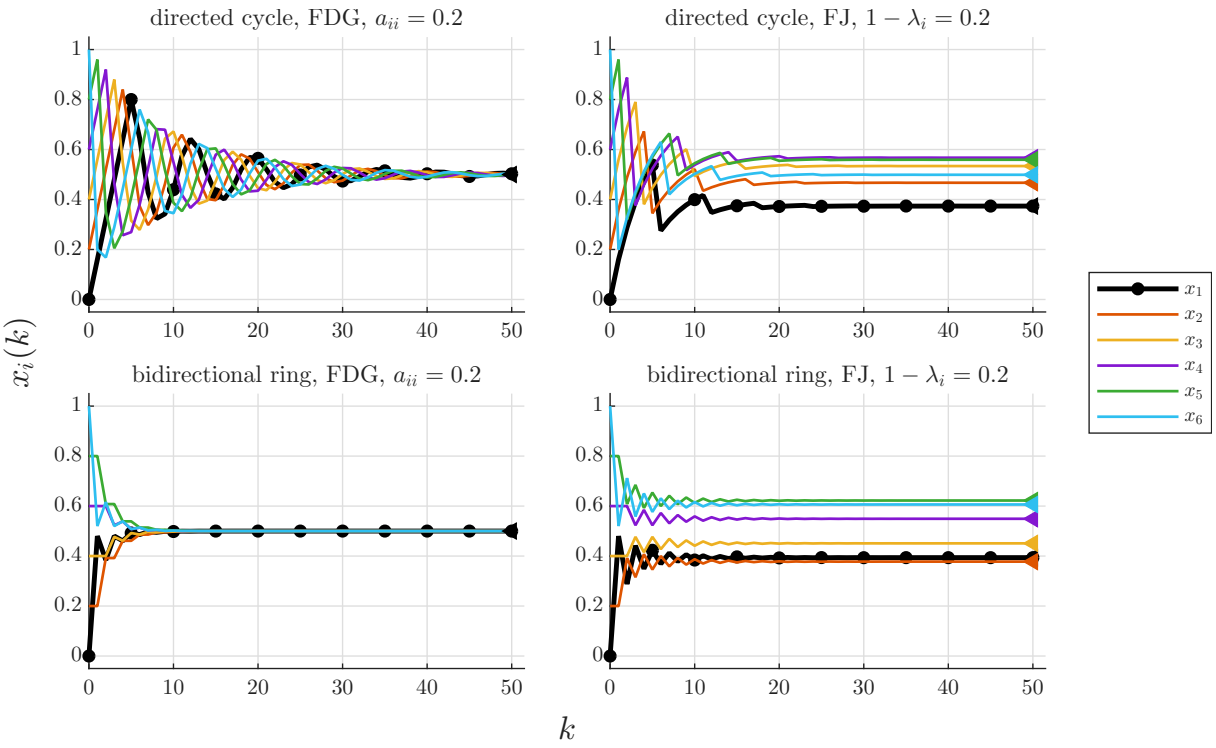


Figure 10: Opinions $x_i(k)$ when every agent gives weight 0.2 to its own opinion, with $x(0) = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$ and $K = 50$. Left: FDG model with $A = 0.2\,I + 0.8\,W$ (self-weight $a_{ii} = 0.2$, current opinion); right: FJ model with $\Lambda = 0.8\,I$ and $u = x(0)$ (attachment $1 - \lambda_i = 0.2$, initial opinion). Top: directed cycle; bottom: bidirectional ring. The triangles mark the limits: the average 0.5 of the initial opinions for the FDG model, and Vu of (9) for the FJ model.

In the FDG model, A is primitive (Corollary 1.14) and doubly-stochastic, so the group agrees on the average 0.5 of the initial opinions (Corollary 2.2): the self-weights slow the convergence down, but do not prevent the consensus. In the FJ model every agent is prejudiced ($\Lambda < I_6$), so the model is asymptotically stable (Corollary 3.3) and $x(\infty) = Vu$, with $V = 0.2\,(I - 0.8\,W)^{-1}$. The matrix V is stochastic (Theorem 3.2): every final opinion is a convex combination of the prejudices, and v_{ij} is the weight of u_j in $x_i(\infty)$. No column of V is zero, so every prejudice enters every final opinion, but with weights that change from row to row: the final opinions are closer to each other than the initial ones, but different~~. A consensus requires equal prejudices: since V is invertible and $V\mathbf{1}_6 = \mathbf{1}_6$, $Vu = c\,\mathbf{1}_6$ only for $u = c\,\mathbf{1}_6$.~~ : the prejudices differ, so by Section 3.1.3 no consensus is possible.

This is the main difference between the two models. On a strongly connected network the FDG model can only reach a consensus or not converge (cases (i)–(ii) of Table 1), since a disagreement needs at least two closed groups (Theorem 2.3); the FJ model disagrees on the same network, so its disagreement comes from the prejudices. In the augmented graph (Section 3.1.3) the six prejudices are six stubborn virtual agents with different opinions, as in Remark 2.6. In social terms, inertia gives way in the end, an anchor does not: a group anchored to different opinions settles on a persistent disagreement, a cleavage of opinions in the words of [5].